



西北工业大学
NORTHWESTERN POLYTECHNICAL UNIVERSITY

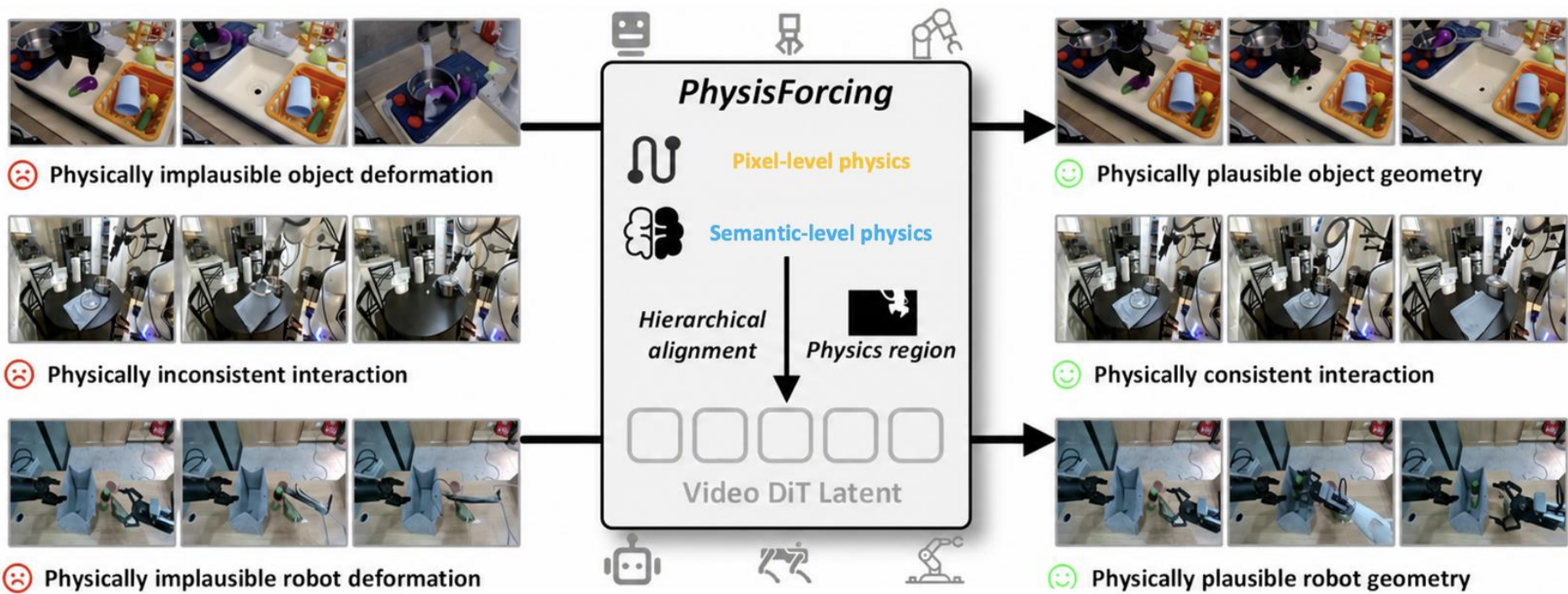
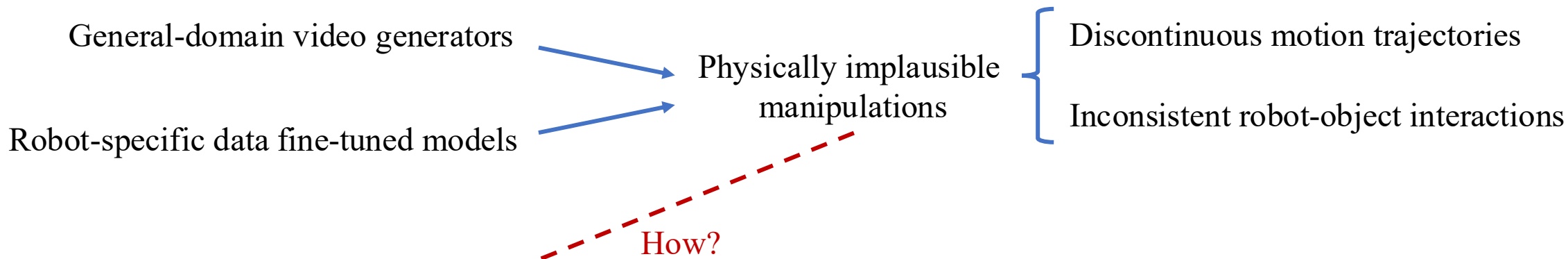
PhysisForcing: Physics Reinforced World Simulator for Robotic Manipulation



Zhengwei Yang

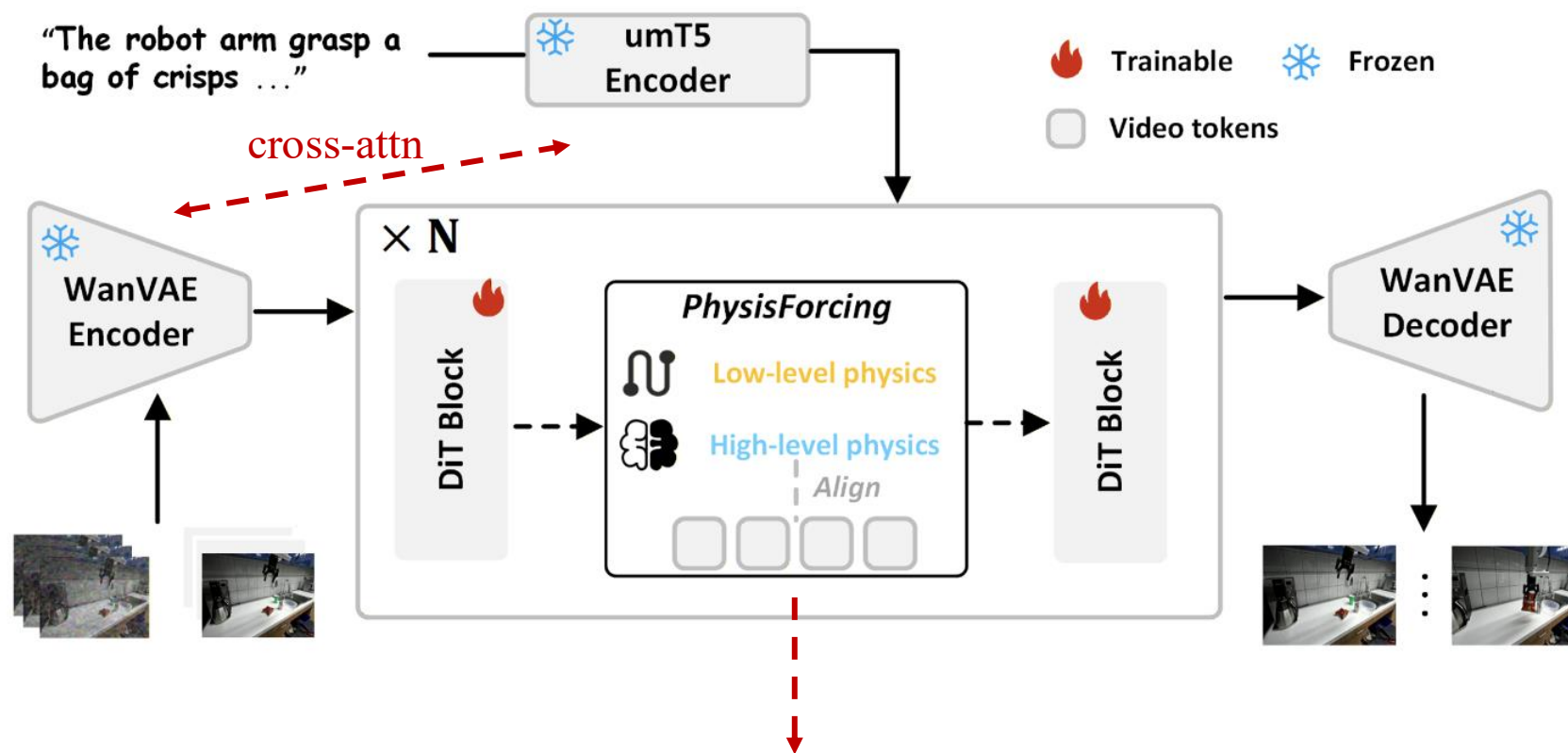
2026.08.09

Motivation



(a) Overview of PhysisForcing

Architecture

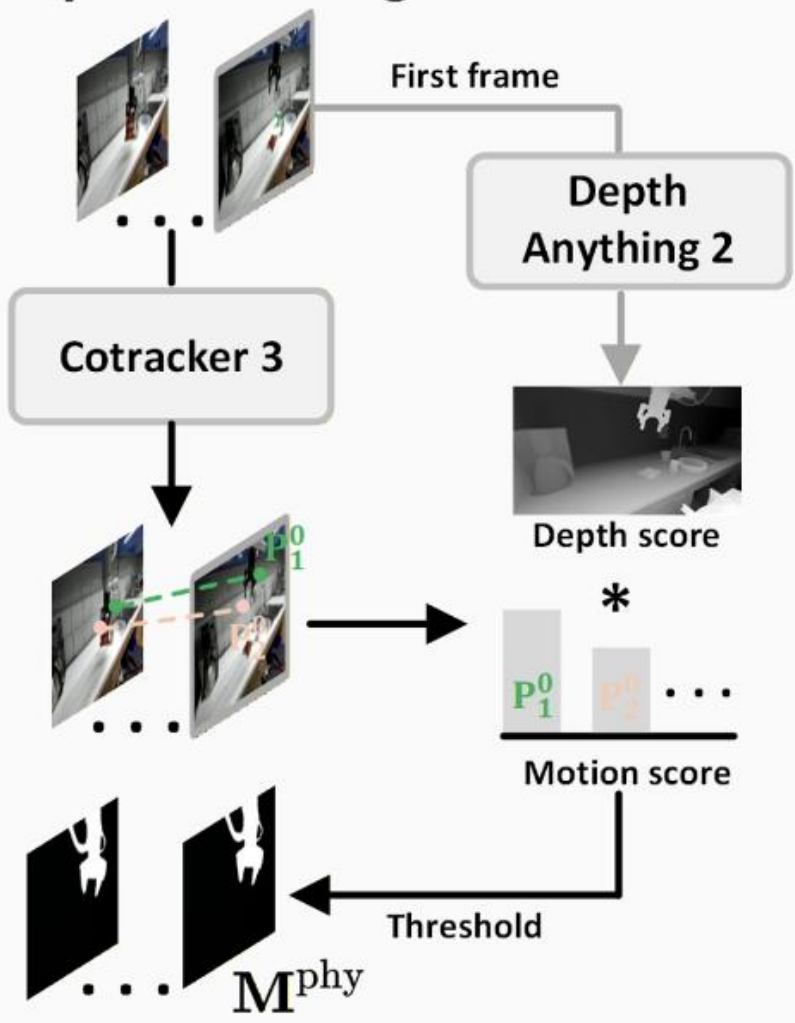


The ℓ -th layer that carry the most informative motion and structure cues.

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{pix}} \mathcal{L}_{\text{pix}}^{\text{phy}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}}^{\text{phy}}$$

Method

Physics-rich region focus



Trajectories $\mathcal{P} = \{\mathbf{p}_i^{1:T}\}_{i=1}^N$

Local Motion $a_i = \sum_{t=1}^{T-1} \|\mathbf{p}_i^{t+1} - \mathbf{p}_i^t\|_2$

$q_i = a_i \cdot r_i$

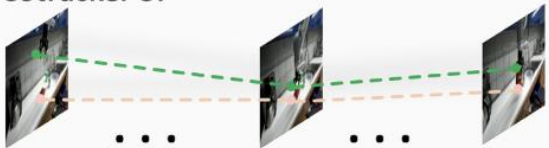
Foreground Weight $r_i = \frac{1}{D_0(\mathbf{p}_i^0) + \epsilon}$

Motion Mask $\mathbf{M}_i^{phy} = \mathbb{I}\left(q_i \geq \frac{1}{N} \sum_{j=1}^N q_j\right) \rightarrow \{0,1\}^{T \times H \times W}$

Method

Pixel-level physics alignment

Cotracker 3:



Similarity-based match:



GT tracks
 $\mathcal{P}^{\text{gt}} \in \mathbb{R}^{K \times T \times 2}$

MSE
loss

Pred tracks
 $\mathcal{P}^{\text{pred}} \in \mathbb{R}^{K \times T \times 2}$

Projected DiT Feature

$$\phi(\mathbf{H}^l) \rightarrow \hat{\mathbf{F}} \in \mathbb{R}^{T \times C \times H \times W}$$

Feature Map

$$\mathbf{Q} = \hat{\mathbf{F}}_0, \quad \mathbf{K}_t = \hat{\mathbf{F}}_t, \quad t = 1, \dots, T-1.$$

Similarity Map

$$\mathbf{s}_i^t(\mathbf{x}) = \frac{\mathbf{Q}(\mathbf{p}_i^0)^\top \mathbf{K}_t(\mathbf{x})}{\sqrt{C}}$$

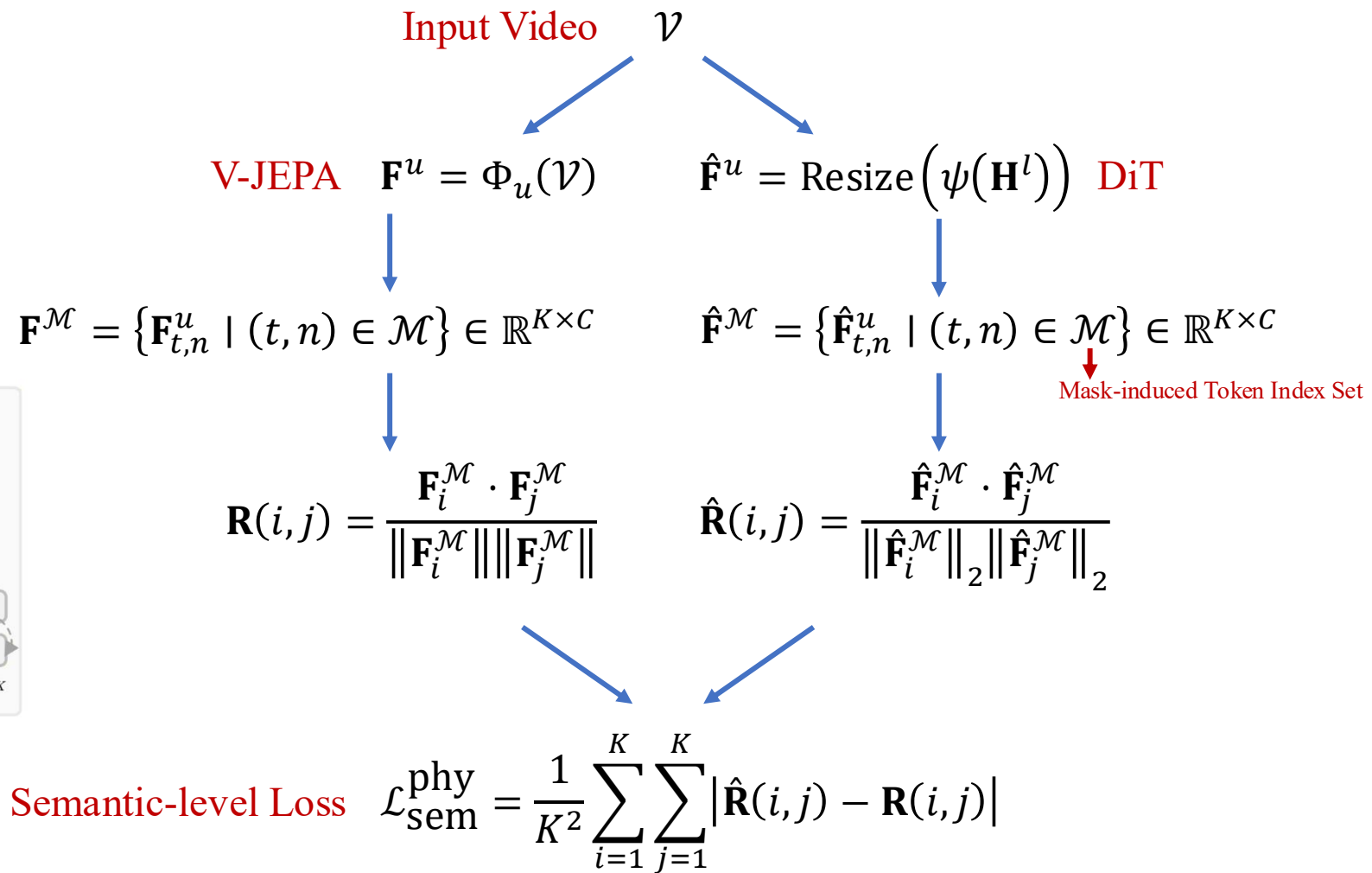
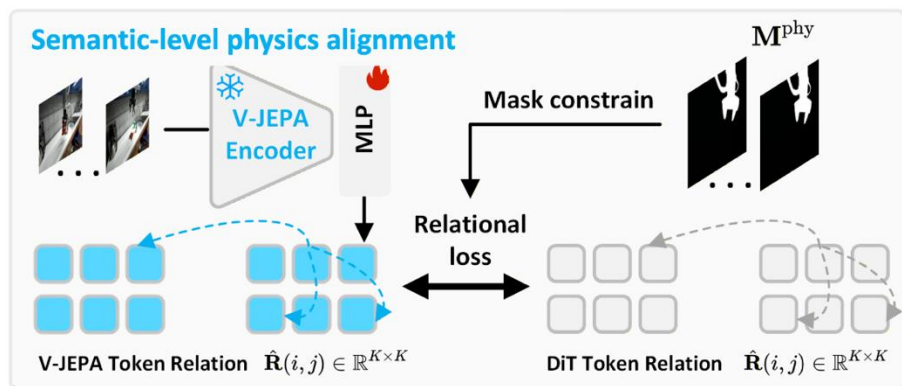
Predicted Point Location

$$\hat{\mathbf{p}}_i^t = \sum_{\mathbf{x} \in \Omega} \text{Softmax}_{\mathbf{x}}(\mathbf{s}_i^t(\mathbf{x})) \mathbf{x}$$

Pixel-level Loss

$$\mathcal{L}_{\text{pix}}^{\text{phy}} = \frac{1}{|\mathbf{M}^{\text{phy}}|} \left\| \mathbf{M}^{\text{phy}} \odot (\mathcal{P}_{\text{pred}} - \mathcal{P}_{\text{gt}}) \right\|_2^2$$

Method



Results

Models	Avg.	Tasks					Embodiments			
		Manipulation	Spatial	Multi-entity	Long-horizon	Reasoning	Single arm	Dual arm	Quadruped	Humanoid
Open-source										
HunyuanVideo 1.5 [47]	46.0	44.2	31.6	31.2	43.8	36.4	51.3	52.6	63.4	59.5
LongCat-Video [43]	43.7	37.2	31.0	22.0	38.4	18.6	58.6	57.6	68.1	62.1
Wan2.1-14B [45]	39.9	34.4	26.8	28.2	33.5	20.5	46.4	49.7	59.5	59.9
LTX-2 [20]	38.1	28.4	30.4	23.3	38.6	16.4	45.3	42.4	62.2	55.5
Wan2.2-Ti2V-5B [45]	38.0	33.1	31.3	14.2	31.8	23.4	43.6	44.8	59.0	60.7
SkyReels [8]	36.1	20.3	27.6	20.3	25.4	23.4	50.7	47.7	58.6	50.9
LTX-Video [19]	34.4	30.2	17.6	21.0	28.0	24.1	44.0	45.6	52.6	46.4
FramePack [53]	33.9	20.6	25.8	17.3	16.9	17.0	44.0	46.4	62.6	54.8
HunyuanVideo [28]	30.3	17.7	18.0	10.8	14.7	3.5	45.4	48.0	62.5	52.4
CogVideoX_5B [50]	25.6	11.6	11.2	9.8	21.2	7.9	33.8	38.5	46.5	49.6
Commercial										
Wan2.6 [45]	60.7	54.6	65.6	47.9	51.4	53.1	66.6	68.1	72.3	66.7
Veo 3.1 [17]	59.9	54.1	47.4	53.4	59.2	46.7	67.0	66.6	74.3	70.4
Seedance 1.5 Pro [38]	58.4	57.7	49.5	48.4	57.0	47.0	64.8	64.1	68.0	69.2
Wan2.5 [45]	57.0	52.7	57.6	40.2	49.6	43.7	68.0	63.4	72.6	65.4
Hailuo v2 [21]	56.5	56.0	63.7	38.6	54.5	47.4	59.4	61.1	64.0	63.5
Veo 3 [17]	56.3	52.1	50.8	43.0	53.0	50.4	63.4	61.0	68.9	63.7
Seedance 1.0 [16]	55.1	54.2	42.5	44.8	45.4	44.2	62.2	64.1	69.8	68.6
Kling 2.6 Pro [27]	53.4	52.9	59.8	36.4	53.0	35.8	57.0	60.5	63.7	61.3
Sora v2 Pro [35]	36.2	20.8	26.8	18.6	25.5	11.5	47.6	51.3	66.4	56.1
Sora v1 [34]	26.6	15.1	22.3	11.1	16.6	13.9	31.4	32.4	54.4	41.9
Robotics-specific										
Cosmos3-Super [1]	58.1	48.7	64.2	44.4	59.1	39.5	61.5	62.3	73.9	69.1
Abot-PhysWorld [11]	52.9	48.6	54.8	43.4	52.3	45.4	66.2	66.8	53.1	45.5
Cosmos 2.5 [1]	46.4	35.8	33.8	20.1	49.6	39.9	54.4	56.0	65.8	62.6
DreamGen(gr1) [24]	42.0	31.2	37.2	29.7	33.4	21.5	56.4	53.2	57.9	57.5
DreamGen(droid) [24]	40.5	35.8	34.8	21.4	31.6	33.9	49.9	47.6	54.2	55.6
Vidar [15]	20.6	7.3	10.6	5.0	5.4	5.0	38.2	41.0	37.4	35.7
UnifoLM-WMA-0 [44]	12.3	3.6	4.0	1.8	6.2	0.0	26.8	19.4	29.3	20.0
Finetuned methods										
Wan2.2-Ti2V-5B [45]	38.0	33.1	31.3	14.2	31.8	23.4	43.6	44.8	59.0	60.7
Wan2.2-Ti2V-5B (ft)	44.8	39.6	41.5	24.9	42.4	28.8	49.2	52.7	61.4	62.6
+ PhysisForcing	47.5	43.4	42.6	29.6	47.5	31.2	51.6	57.4	59.7	64.2
Wan2.2-I2V-A14B [45]	50.7	38.1	45.4	37.3	50.1	33.0	60.8	58.2	69.0	64.8
Wan2.2-I2V-A14B (ft)	57.9	52.3	62.8	45.2	54.5	47.8	64.2	63.5	65.6	65.3
PF-Wan	62.0	56.4	65.4	49.1	58.4	52.4	68.7	69.6	69.2	68.5
Cosmos3-Nano [1]	58.4	55.0	67.0	46.6	57.6	39.4	59.1	61.1	73.1	66.6
Cosmos3-Nano (ft)	61.5	57.8	67.4	49.2	59.3	48.5	66.5	67.1	70.6	67.1
PF-Cosmos	63.8	58.9	69.7	51.3	60.8	53.0	69.3	70.0	72.2	69.0

R-Bench Evaluation

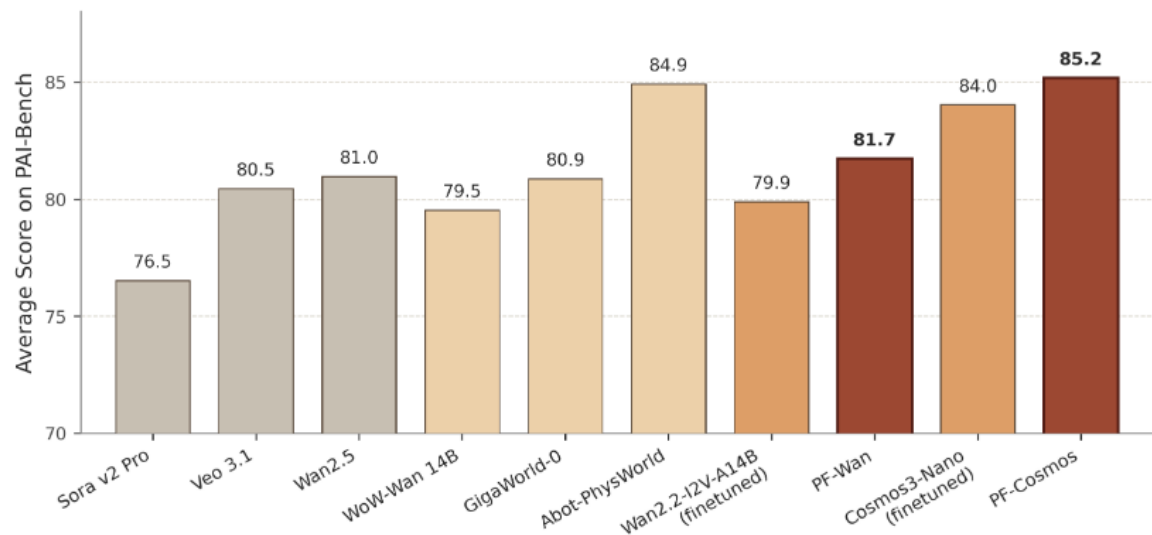
P.S. 650 image-text pairs across task-oriented and embodiment-specific dimensions.



The robotic hand moves the red apple onto the second-level wooden platform.

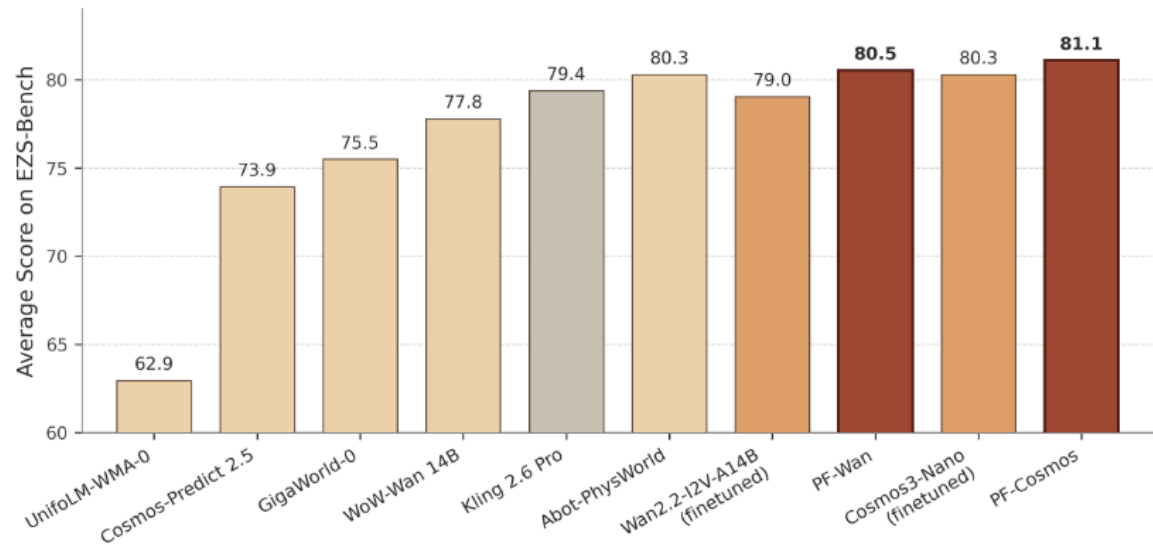
Qualitative Comparison with SOTA

Results



PAI-Bench Evaluation

P.S. 174 [real-world](#) image-prompt pairs



EZS-Bench Evaluation

P.S. A training-independent zero-shot benchmark of 196 unseen robot-task-scene combinations probing [out-of-distribution generalization](#).

Results

Table 2 Downstream policy success rate (% , 200 rollouts/task) on RoboTwin 2.0 (Fast-WAM backbone).

Task	Baseline	PhysisForcing	Δ
<i>place_empty_cup</i>	41.5	63.0	+21.5
<i>press_stapler</i>	49.0	60.0	+11.0
<i>grab_roller</i>	58.5	63.0	+4.5
<i>shake_bottle</i>	97.5	94.5	-3.0
<i>adjust_bottle</i>	93.0	93.0	0.0
<i>stack_bowls_two</i>	69.5	63.0	-6.5
Average	68.2	72.8	+4.6

Fast-WAM with

Wan2.2-TI2V-5B trained with PhysisForcing

Table 3 WorldArena action planner closed-loop success rate (%); best world model in **bold**.

Model	Task 1	Task 2	Avg.
Genie Envisioner [32]	10.0	20.0	15.0
Tesseract [56]	1.0	35.0	18.0
RoboMaster	8.0	20.0	14.0
Vidar [15]	2.0	19.0	10.5
WoW [12]	20.0	21.0	20.5
Wan2.2-TI2V-5B	12.0	20.0	16.0
+ PhysisForcing	22.0	26.0	24.0

Frame prediction + IK dynamics

Ablation Study

Model	Emb.	Tasks	Avg.
Wan2.2-TI2V-5B (ft)	56.5	35.4	44.8
+ $\mathcal{L}_{pix}^{phy}$	59.0	37.8	47.2
+ $\mathcal{L}_{sem}^{phy}$	58.4	36.5	46.2
+ PysisForcing	58.2	38.9	47.5
Wan2.2-I2V-A14B (ft)	64.7	52.5	57.9
+ $\mathcal{L}_{pix}^{phy}$	67.5	55.2	60.7
+ $\mathcal{L}_{sem}^{phy}$	66.8	54.6	60.0
+ PysisForcing	69.0	56.3	62.0

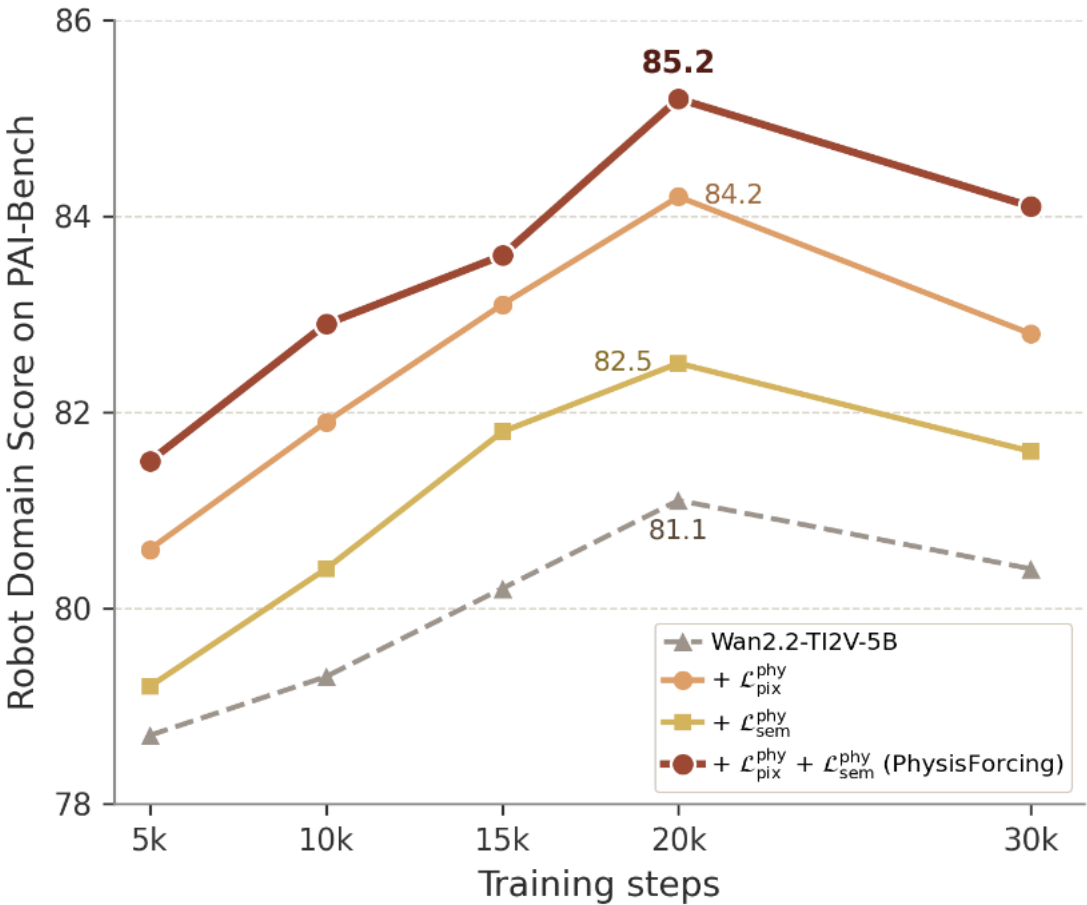
Per-component ablation on R-Bench

Model	Emb.	Tasks	Avg.
Wan2.2-TI2V-5B (ft)	56.5	35.4	44.8
w/o Physics region focus	57.0	37.2	46.0
w/ Physics region focus	58.2	38.9	47.5

Physics region focus ablation on R-Bench

Layer	10	15	20	25
Robot Domain Score	83.9	85.2	84.1	83.2

Alignment-block (layer index) ablation



Reflection

Physical plausibility
any fine-tuning method can reach

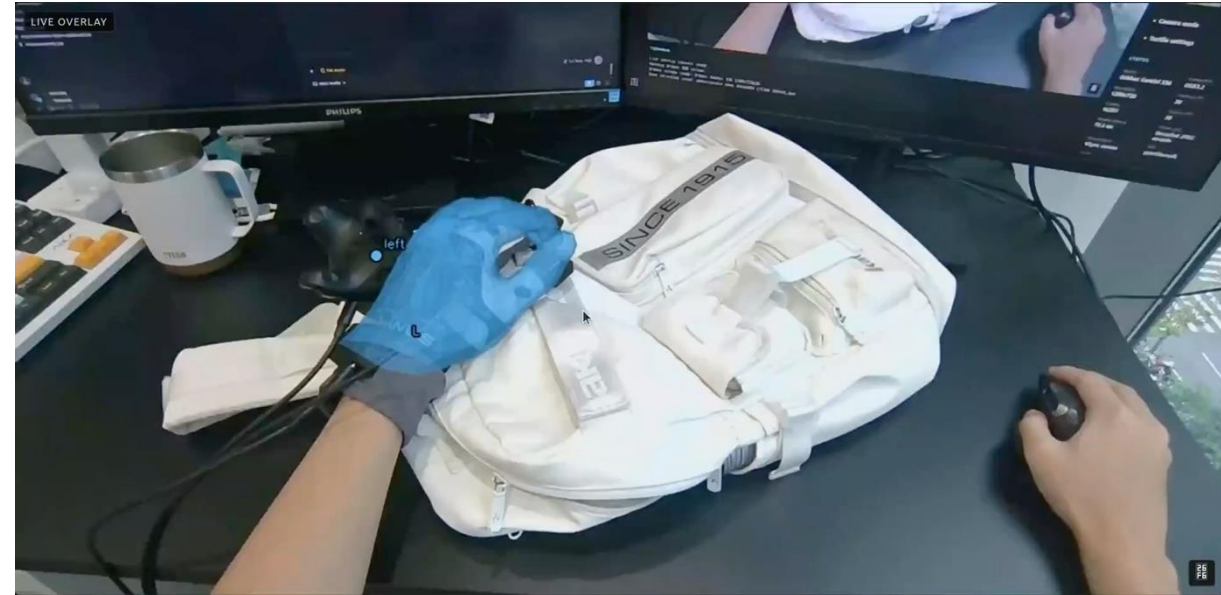


World knowledge and long-horizon
temporal reasoning ability
of
current open-source video generators

HumanTouch

Scaling Whole-Hand Tactile Manipulation
with Human Data

X SPARK AI



<https://xsparkai.com/sparklab/humantouch/>



西北工业大学
NORTHWESTERN POLYTECHNICAL UNIVERSITY

Thanks for listening!

Zhengwei Yang

2026.08.09